

ARCHANA NAGARAJ

Bioinformatics Engineer · AI / ML

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PROFESSIONAL SUMMARY

Bioinformatics engineer who built and independently deployed a production-grade DNA Sequence Analysis Platform (Next.js, FastAPI, Neon, Vercel) covering GC/AT composition, RNA transcription, protein translation, batch CSV ingestion, and automated PDF reporting. Combines deep computational biology knowledge — NGS, variant analysis, protein structure, molecular docking — with practical AI/ML and full-stack skills. Looking for roles where biology, software, and data science.

EXPERIENCE

Bioinformatics Research Collaborator · Independent Projects & Research | Remote | 2024 – Present

- Reviewed and synthesized primary research literature on genomic sequencing, NGS pipelines, and computational biology methods to guide platform development.
- Designed and independently built a full-stack DNA Sequence Analysis Platform (Next.js, TypeScript, FastAPI, Neon/PostgreSQL) — all architecture decisions made solo.
- Implemented computational biology workflows including GC/AT composition, nucleotide analysis, reverse complement, RNA transcription, codon to amino acid translation, reference similarity scoring and batch sequence analysis.
- Built a CSV batch ingestion pipeline for high-throughput multi-sample analysis workflows.
- Deployed on Vercel with GitHub Actions CI/CD; live DNA Sequence : <https://dna-sequence-analyzer-two.vercel.app/>
- Implemented automated PDF report generation for single-sequence and batch comparison outputs.

PROJECTS

DNA Sequence Analysis Platform | Next.js · TypeScript · FastAPI · Neon · PostgreSQL · Vercel . —> <https://dna-sequence-analyzer-two.vercel.app/>

- Production-deployed full-stack bioinformatics web app: composition analysis, transcription, translation, batch CSV processing, reference comparison (MATCH/MISMATCH), and PDF exports.
- Serverless PostgreSQL backend via Neon; CI/CD via GitHub Actions + Vercel.

Banana Leaf Nutrient Deficiency Detection | Python · TensorFlow · CNN · Image Augmentation

- Trained a 9-class CNN on banana leaf imagery with preprocessing and augmentation; analyzed failure modes and identified EfficientNet transfer learning as the path to improve beyond 60% baseline.
- Designed for real-time deployment on agricultural edge devices.

Diabetes Risk Prediction | Python · Scikit-Learn · Logistic Regression

- Binary classifier (75% accuracy); applied SHAP values to surface top clinical features (glucose, BMI, age) for interpretable, clinically contextualized outputs.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C, HTML/CSS

AI / ML: PyTorch, TensorFlow/Keras, CNN, Transformers, DNA-BERT, Scikit-Learn, Generative AI

Bioinformatics: FastQC, SAMtools, BLAST, Clustal Omega, AlphaFold, PyMOL, AutoDock Vina, Discovery Studio, UniProt, PyRx

Full-Stack: Next.js, React, FastAPI, Node.js, PostgreSQL (Neon), MongoDB, REST APIs, Pandas, NumPy

DevOps / Cloud: AWS, Vercel, Git/GitHub, GitHub Actions (CI/CD), Google Colab, Jupyter, VS Code

EDUCATION

B.Sc. Bioinformatics · REVA University, Bengaluru | 2021–2024 | GPA: 3.35 / 4.0

- Relevant coursework: Computational Biology, NGS Data Analysis, Structural Bioinformatics, Molecular Docking, Machine Learning for Biological Data, DBMS, Genomics and Proteomics

CERTIFICATIONS & TRAINING

- Bioinformatics Algorithms and Its Application — NPTEL
- Genomic Data Science Specialization — Johns Hopkins / Coursera (in progress)
- AWS Cloud Practitioner Essentials — AWS skill Builder (in progress)